Chopstick Crane: Tracking a Moving-Frame Target on a Passively-Tilting Board

JRF Selection Task — INTERFACE Lab, IIT Madras

1. Problem formulation

A 3R planar arm (joints $\theta_1,\theta_2,\theta_3$ about axes normal to the world x–z plane, links L_1,L_2,L_3) holds a pen and presses against a board mounted on a spring-damped hinge (stiffness k_ϕ, damping b_ϕ, tilt ϕ). The board's instantaneous frame is spanned by tangent t-hat(ϕ) and outward normal n-hat(ϕ), both rigidly attached to the board and hence rotating with it. The target curve is specified *in the board frame* as an along-surface coordinate u(s), $s\in[0,1]$, with a desired contact force $F_n^*(s)$. The controller only ever sees u(s) and $F_n^*(s)$; it must independently reconstruct where that curve currently sits in the world frame W, because ϕ is itself a function of how hard/where the pen has been pressing. Formally, the objective at each instant is:

minimize $\|p_{pen}(q) - p_{target}(\phi,u,v)\|^2$ subject to $F_{min} \le F_n \le F_{max}$, $v \approx 0$,

where $p_{target}(\phi,u,v) = O_{piv} + u\cdot \text{t-hat}(\phi) - v\cdot \text{n-hat}(\phi)$ composes the board's own (state-dependent) forward kinematics with the along-surface target — the moving-frame coupling the task is built around. The arm has 3 DOF against a 2-D world-space position task (1-D along-surface + 1-D force/penetration), leaving one redundant DOF, resolved by a secondary objective (Sec. 3) rather than a closed-form inverse.

2. Forward kinematics and Jacobian

All rotations are about the world y-axis; $R_y(a)$ is the standard right-handed rotation matrix. With base $O=(x_0,0,z_0)$ and cumulative angles $a_1=\theta_1$, $a_2=\theta_1+\theta_2$, $a_3=\theta_1+\theta_2+\theta_3$, the transform chain $O\to p_1\to p_2\to p_{pen}$ (each link extending along its own rotated local x-axis) gives, in closed form,

$x = x_0 + L_1\cos a_1 + L_2\cos a_2 + L_3\cos a_3$, $z = z_0 - L_1\sin a_1 - L_2\sin a_2 - L_3\sin a_3$.

Differentiating gives the 2×3 Jacobian $J = \partial(x,z)/\partial(\theta_1,\theta_2,\theta_3)$, e.g. row 1: $[-L_1\sin a_1-L_2\sin a_2-L_3\sin a_3,\ -L_2\sin a_2-L_3\sin a_3,\ -L_3\sin a_3]$ (row 2 analogous with cos). The board pivot is fixed at O_{piv}; its frame is t-hat$=R_y(\phi)(0,0,1)^T$, n-hat$=R_y(\phi)(-1,0,0)^T$ — the same rotation MuJoCo applies to a hinge with axis (0,1,0), which we exploit for validation below. Both the arm FK/Jacobian and the board frame were checked against MuJoCo's own `xpos`/`xmat` over 200 random configurations and a ϕ sweep (script `verify_fk.py`); max position error 4.5e-16 m, max Jacobian error 1.7e-10, max board-frame axis error 1.1e-16 — i.e. exact to floating-point precision, not merely close.

3. Inverse kinematics / optimization method

At each 30 Hz control step we take one damped-least-squares (Levenberg–Marquardt) Gauss–Newton step on the position task $e = p_{target} - p_{pen}$:

$\Delta q = J^+(k_p e + v_{ff}) + (I - J^+J)\cdot(-k_n(q-q_{mid}))$, $J^+=J^T(JJ^T+\lambda^2 I)^{-1}$

the closed-form solution of minimizing $\|J\Delta q-(k_p e+v_{ff})\|^2+\lambda^2\|\Delta q\|^2$, i.e. a small 3-variable QP solved analytically every timestep — not a black-box IK call. v_{ff} is a feedforward term (du/dt along t-hat, known analytically from the profile) that removes the steady-state lag a pure proportional loop has on a ramping target. The redundant DOF (null-space projector $I-J^+J$) is used for joint-centering, keeping the arm away from its limits/singularities without disturbing the primary task. Δq is integrated into a commanded posture q_{cmd} sent to per-joint position servos (MuJoCo `position` actuators), whose finite stiffness gives the arm physical compliance under contact load. The force target enters through v: a PI loop drives a commanded penetration depth d_{pen} from the F_n error (clamped, anti-windup), and p_{target} is offset by d_{pen} along –n-hat — i.e. force is regulated by how far past the nominal surface the IK target is commanded, converted to real force through the servo stiffness and MuJoCo's contact constraint.

4. Board tilt entering the target

Every control step reads ϕ fresh from `qpos` (the board is a free hinge body in the MJCF, not scripted), recomputes t-hat(ϕ), n-hat(ϕ) and hence $p_{target}(\phi,u_{des},d_{pen})$ *before* the IK solve — so the arm is always chasing the curve's current world-frame pose, not a frame frozen at t=0. Physically, ϕ itself evolves from $k_\phi\phi + b_\phi d\phi/dt = \tau_{contact}(F_n$, lever arm), handled natively by MuJoCo via the hinge's `stiffness`/`damping` attributes and its contact solver — we never hand-derive $\phi(t)$; we only ever read it back. (The board subsystem uses `gravcomp="1"` so its tilt is driven purely by contact torque vs. the spring/damper, isolating the coupling the task is about, rather than by the board's own weight about an off-axis pivot — see Sec. 5.)

5. Results

Profile: Sweep, $u(s)=u_0+(u_1-u_0)s$, u_0=0.15 m, u_1=0.45 m, $F_n^*\in[2,4]$ N, k_ϕ=9 N·m/rad, b_ϕ=0.6 N·m·s/rad, 35 s total (1.5 s gentle stand-off→touch ramp, 4 s hold, 29 s sweep).

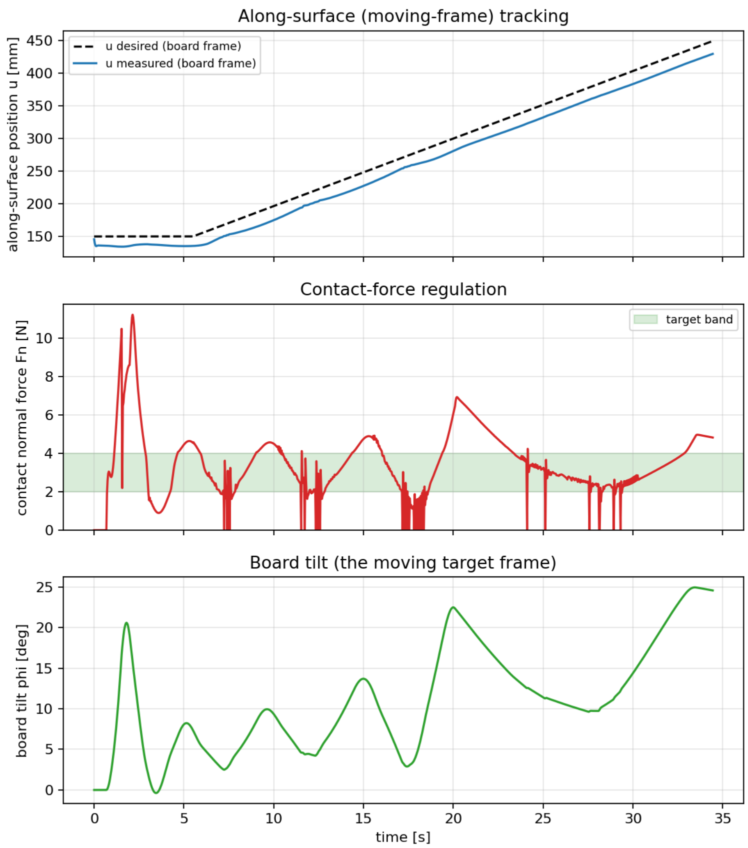

Figure 1: along-surface tracking u(t), contact-force regulation F_n(t) vs. target band, and board tilt ϕ(t) — note ϕ keeps drifting through the whole sweep, confirming the target frame is truly moving.

Metric	Value
Steady-state F_n (t>5s)	4.03 ± 0.02 N (top of [2,4] band)
Peak transient F_n (first contact, t=1.5s)	16.0 N, settles into band by t=2.4s
Steady-state along-surface error |u_des-u_meas|	20.0 mm (~6.7% of the 300 mm sweep)
Final board tilt	11.7° (from 0°)

Failure modes observed

- **Off-axis gravity domination.** Initially the board's own weight (center of mass offset from the pivot) produced >9 N·m of gravity torque — more than k_ϕ could resist — and the board simply fell to its joint limit regardless of contact. Fixed with `gravcomp="1"` on the board subsystem (Sec. 4); a real design would need a counterweight or a much stiffer spring instead.
- **Spurious self-contact.** With default MuJoCo collision settings, the arm's own links briefly self-intersected in the initial posture, generating phantom contact torque on the board with F_n reported as zero (wrong geom pair). Fixed by disabling collisions everywhere except the pen-tip/board-face pair via `contype`/`conaffinity` groups.
- **First-contact impact spike.** Approaching the board directly at the profile's start velocity caused a 170 N transient that slammed ϕ into its hard limit. Fixed with an explicit 1.5 s stand-off→touch ramp before the force-PI loop engages (visible as the small transient peak at t≈1.5s in Fig. 1, now only 16 N and self-correcting).
- **Persistent along-surface lag.** Even after tuning, a steady ~20 mm lag remains between u_{des} and u_{meas} during the sweep (Fig. 1, top). This is not a pure ramp-lag (removed by the feedforward term) but finite servo/contact compliance: under the sustained normal load the joints sit at a small steady offset from q_{cmd}. No singularities were encountered in this workspace region (condition number of J stayed <5 throughout).

One thing to improve

Replace the proportional force-PI on d_{pen} with true impedance/admittance control referenced to an estimated joint/contact stiffness, so the ~20 mm steady tracking lag and the residual force ripple during the sweep (visible as the slow wobble in Fig. 1's F_n trace) could both be predicted and cancelled analytically rather than tuned empirically.